



STARSYN AI · ACADEMY

Course *Catalogue*

Thirty-eight courses across five families — from a single day on the plant floor to a six-month career track. Every one capped at eight seats.

38

COURSES

5

FAMILIES

6

AGENT
COURSES

8

SEATS EACH

Everything we teach

Five families. The first two get a department capable; the third and fourth make builders; the fifth is where somebody changes career. Every course draws on the same sixteen-module curriculum, so a learner can move between them without repeating themselves.

A · Applied AI programmes

No.	Course	Duration	Level
01	Applied AI for Professionals	4 weeks · 24 hours	Beginner to intermediate
02	Applied AI for Business Owners & Managers	3 weeks · 18 hours	Beginner to intermediate
03	Applied AI for Manufacturing, EHS & Quality	5 weeks · 30 hours	Beginner to intermediate
04	Applied AI for Teachers & Schools	5 weeks · 30 hours	Beginner to intermediate
05	Applied AI for HR Teams	4 weeks · 24 hours	Beginner to intermediate
06	Applied AI for Finance & Accounts	4 weeks · 24 hours	Beginner to intermediate
07	Applied AI for Sales & Marketing	4 weeks · 24 hours	Beginner to intermediate
08	Applied AI for Students	6 weeks · 36 hours	Beginner to intermediate
09	Applied AI for Government & Compliance	4 weeks · 24 hours	Beginner to intermediate

B · Core skill courses

No.	Course	Duration	Level
10	Prompt Engineering	2 weeks · 12 hours	Intermediate
11	AI Automation with n8n	3 weeks · 18 hours	Intermediate
12	Python for AI	4 weeks · 24 hours	Intermediate
13	Data Analysis with AI	3 weeks · 18 hours	Intermediate
14	Machine Learning	6 weeks · 36 hours	Intermediate
15	Deep Learning	6 weeks · 36 hours	Advanced
16	Natural Language Processing	5 weeks · 30 hours	Advanced
17	Computer Vision	5 weeks · 30 hours	Advanced
18	Generative Media	3 weeks · 18 hours	Intermediate

Agents, advanced work and flagships

C · The agent track

No.	Course	Duration	Level
19	AI Agents — Foundations	3 weeks · 18 hours	Intermediate
20	Agentic Automation for Business	4 weeks · 24 hours	Intermediate
21	Agent Engineering	8 weeks · 48 hours	Advanced
22	Agent Evaluation & Guardrails	4 weeks · 24 hours	Advanced
23	Agent Operations in Production	4 weeks · 24 hours	Advanced
24	Build an Agent Product	8 weeks · 48 hours	Advanced

D · Advanced & specialist

No.	Course	Duration	Level
25	RAG Engineering	6 weeks · 36 hours	Advanced
26	LLM Engineering	10 weeks · 60 hours	Advanced
27	AI Backend & APIs	5 weeks · 30 hours	Advanced
28	MLOps & Deployment	6 weeks · 36 hours	Advanced
29	AI Evaluation Engineering	4 weeks · 24 hours	Advanced
30	AI Security	4 weeks · 24 hours	Advanced
31	Document Intelligence	5 weeks · 30 hours	Advanced
32	Time Series & Forecasting	5 weeks · 30 hours	Advanced
33	AI Governance for Indian Companies	3 weeks · 18 hours	Intermediate
34	AI Product Management	4 weeks · 24 hours	Intermediate

E · Flagship programmes

No.	Course	Duration	Level
35	Applied AI Build Cohort	4 weeks · 16 hours live	All levels
36	AI Career Track	6 months · 480 hours	Beginner to advanced
37	LLM & Agent Engineering Track	12 weeks · 84 hours	Advanced
38	Corporate Custom Programme	Scoped per client	Mixed

A

Applied AI programmes

Three to six weeks, built around a job rather than a technology, and technical from the first session. Every one carries the same mandatory foundation — AI terminology, Python, data handling and machine learning basics — then goes domain-specific, and every one ends with the participant having built a working agent on their own data. No prerequisites, but expect real homework.

COURSES IN THIS FAMILY

01 · Applied AI for Professionals

02 · Applied AI for Business Owners & Managers

03 · Applied AI for Manufacturing, EHS & Quality

04 · Applied AI for Teachers & Schools

05 · Applied AI for HR Teams

06 · Applied AI for Finance & Accounts

07 · Applied AI for Sales & Marketing

08 · Applied AI for Students

09 · Applied AI for Government & Compliance

MANDATORY FOUNDATION, CARRIED BY EVERY APPLIED PROGRAMME

AI · ML · DL · GenAI taxonomy

Tokens & embeddings

Context windows

Hallucination and its causes

Python essentials

NumPy & pandas

Train / test splits

Overfitting & evaluation

Regression & classification

Prompt schemas & few-shot

Embeddings & RAG

Tool calling & the agent loop

n8n automation

Human checkpoints

DPDP & data handling

Nobody leaves an applied programme having only learned to prompt a chatbot. Every participant writes Python, touches real data, trains a simple model, and ships an agent built on their own work.

01

Applied AI for Professionals

Not a tour of chat tools. Foundations, Python, retrieval and a working agent that automates one real part of your job.

4 weeks · 24 hours

Online live + labs · 8 seats

Beginner to intermediate

Curriculum modules 01, 03, 04, 07, 08, 09, 12, 13

8 seats

WHO THIS IS FOR

- Managers, engineers, analysts and coordinators who produce documents and reports
- Professionals told to “use AI” and given no technical grounding
- Anyone who wants to build the automation rather than wait for IT

PREREQUISITE

None. A laptop, and one recurring task from your own job.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	AI, ML, DL and GenAI: what each term actually denotes · Tokens, embeddings, context windows, temperature and top-p · Why models hallucinate: next-token prediction, not lookup	Understand
Stage 2	Python essentials: variables, functions, files, JSON, error handling · pandas for real data: load, clean, join, group, export · System prompts, few-shot examples and JSON output schemas	Practise
Stage 3	Supervised learning in one sitting: features, labels, train/test, overfitting · Regression and classification on your own numbers with scikit-learn · Embeddings and vector search: RAG over your own documents	Build
Stage 4	Tool calling and the reason-act-observe agent loop · n8n: triggers, nodes, credentials, error handling, run logs · Human checkpoints, cost per run, and what data never leaves the building	Harden and hand over

BY THE END YOU CAN

- Use the technical vocabulary correctly in front of engineers and vendors
- Write and debug small Python that cleans and analyses your own data
- Build and run an agent that removes one recurring task from your week

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A RAG assistant over your own reports and reference documents, answering with sources
- An agent that reads incoming material, extracts structured fields and files it, pausing for approval on low confidence
- A scheduled n8n workflow that produces one recurring report from raw data

02

Applied AI for Business Owners & Managers

Enough technical depth to judge a proposal, plus a working agent built on your own operations.

3 weeks · 18 hours

Online live + labs · 8 seats

Beginner to intermediate

Curriculum modules 01, 03, 04, 07, 09, 12, 33

8 seats

WHO THIS IS FOR

- Founders and directors deciding where AI money goes
- Department heads accountable for cost, quality or headcount
- Managers evaluating vendors and unable to check their claims

PREREQUISITE

None. Bring your own P&L lines and one broken process.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	The taxonomy: rules, classical ML, deep learning, generative models · What training, fine-tuning, inference and context actually cost · Reading a vendor's architecture claim and finding the gap	Understand
Stage 2	Python and pandas at a level sufficient to check a analyst's work · Classification and forecasting: what a model can and cannot promise · Accuracy, precision, recall and why a 97% claim can be worthless	Practise
Stage 3	RAG versus fine-tuning: which problem each one solves · Agent loops, tool permissions and where autonomy becomes liability · Unit economics: cost per run, per user, per month at real volume	Build
Stage 4	Build versus buy, with total cost of ownership · DPDP obligations, data classification and vendor due diligence · Writing an internal AI usage policy that survives an audit	Harden and hand over

BY THE END YOU CAN

- Interrogate a vendor's technical claim rather than accepting it
- Model the true running cost of an AI feature before approving it
- Deploy one agent into your operations with governance around it

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- An operations agent that monitors a data source and escalates exceptions to a named person
- A costed opportunity map of your business with effort, payback and risk per item
- An internal AI usage and data-handling policy for your own organisation

03

Applied AI for Manufacturing, EHS & Quality

Taught bilingually on the plant floor: foundations, Python, vision, retrieval, and agents built on your own registers.

5 weeks · 30 hours

On site + online labs · 8 seats

Beginner to intermediate

Curriculum modules 01, 03, 04, 07, 08, 09, 11, 12, 13

8 seats

WHO THIS IS FOR

- EHS officers, safety supervisors and section in-charges
- Quality and production engineers with sensor and inspection data
- Teams maintaining statutory registers, returns and audit evidence

PREREQUISITE

Bring last month's registers, logs and reports.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	AI, ML, DL, GenAI and where each fits on a plant floor · Tokens, embeddings, context and hallucination, stated plainly · Python essentials and pandas on your own log exports	Understand
Stage 2	Cleaning sensor and register data: missing values, duplicates, units · Classification for incident severity; regression for consumption · Precision and recall on imbalanced safety data	Practise
Stage 3	Anomaly detection with isolation forests on process data · Computer vision tasks: classification, detection, OCR on registers · System prompts and schemas for audit-grade rewriting	Build
Stage 4	RAG over IS standards, SOPs, circulars and internal procedures · Agents with tool calling: read, check, draft, escalate · n8n triggers, human sign-off gates, and DPDP-safe data handling	Harden and hand over

BY THE END YOU CAN

- Speak accurately about models, data and limits in an audit or management review
- Analyse your own plant data in Python rather than requesting a report
- Run an agent that prepares statutory documentation for a human to sign

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- An EHS agent that classifies raw observations by risk, cites the governing clause and drafts the register entry
- A retrieval system over your own standards that answers with the clause reference
- A monthly EHS or quality pack assembled from source logs, with a sign-off gate

04

Applied AI for Teachers & Schools

Teachers build their own teaching agents — question papers, marking support, records — with the technical grounding to control them.

5 weeks · 30 hours

On site or online + labs · 8 seats

Beginner to intermediate

Curriculum modules 01, 03, 04, 07, 08, 09, 12, 13

8 seats

WHO THIS IS FOR

- Teachers across subjects, boards and languages
- Coordinators, heads of department and examination cells
- School administrators handling records, circulars and reporting

PREREQUISITE

Bring your syllabus, past papers and one class's marks.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	AI, ML, DL and GenAI: the vocabulary parents will ask about · Tokens, context windows and why a whole textbook will not fit · Temperature, system prompts and few-shot examples	Understand
Stage 2	JSON output schemas for a fixed paper structure · Python and pandas on your marks sheet: spread, weak topics · Item analysis: difficulty index and discrimination	Practise
Stage 3	Clustering students by performance pattern, and its ethical limits · Embeddings and RAG over your syllabus, textbook and past papers · Bloom-level tagging and rubric-based evaluation of generated items	Build
Stage 4	Agents with tool calling: retrieve syllabus, draft, check, format · n8n with Google Classroom, Sheets, Docs, Drive and Gmail triggers · Student data protection, disclosure, and detecting AI-written work	Harden and hand over

BY THE END YOU CAN

- Explain to colleagues and parents what these systems do, correctly
- Analyse your own class data in Python rather than by eye
- Run teaching agents you built yourself, tuned to your syllabus

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A question-paper agent grounded in your syllabus and past papers, with a stated difficulty spread
- A marks-analysis notebook flagging weak topics per student and class
- An n8n workflow that generates, formats and files worksheets or circulars bilingually

05

Applied AI for HR Teams

A policy agent, a hiring pipeline and the statistical literacy to know where automation becomes discrimination.

4 weeks · 24 hours

Online live + labs · 8 seats

Beginner to intermediate

Curriculum modules 01, 03, 04, 07, 08, 09, 12, 13

8 seats

WHO THIS IS FOR

- HR generalists, recruiters and people partners
- Managers handling policy questions and onboarding
- Small companies with no dedicated HR technology

PREREQUISITE

Bring your handbook, JDs and one hiring pipeline.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	The AI taxonomy and what an “AI screening tool” really contains · Tokens, embeddings and semantic similarity between CVs and JDs · Python and pandas on attrition, headcount and pipeline data	Understand
Stage 2	Classification basics, and why a hiring model can encode past bias · Confusion matrices, false negatives and the human cost of each · Fairness, adverse impact and what must stay a human decision	Practise
Stage 3	System prompts and schemas for structured JD and scorecard output · RAG over your handbook so answers quote the actual policy clause · Agent loops with tool calling across ATS, mail and sheets	Build
Stage 4	n8n workflows for onboarding, document collection and reminders · Personal data under DPDP: consent, retention, access, deletion · Audit logging so every automated action can be explained	Harden and hand over

BY THE END YOU CAN

- Challenge a vendor's screening claim on statistical grounds
- Answer routine policy questions instantly with a citation
- Automate onboarding while keeping every decision attributable to a person

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A policy agent grounded in your own handbook, answering with clause citations and refusing when unsure
- An onboarding workflow in n8n that collects documents, tracks completion and escalates
- A pipeline analysis in Python showing where candidates drop out and why

06

Applied AI for Finance & Accounts

Extraction, reconciliation and forecasting done in Python, with an agent that closes the monthly loop.

4 weeks · 24 hours

Online live + labs · 8 seats

Beginner to intermediate

Curriculum modules 01, 03, 04, 07, 09, 12, 13, 31

8 seats

WHO THIS IS FOR

- Accountants, finance executives and controllers
- Business owners running their own books
- Anyone assembling an MIS pack by hand every month

PREREQUISITE

Excel fluency. Bring two sources that must reconcile.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Why arithmetic inside generated prose is unverified · Python and pandas: the reconciliation toolkit end to end · Fuzzy matching and deduplication across two messy ledgers	Understand
Stage 2	OCR and layout parsing for invoices, statements and bills · Structured extraction with JSON schemas and validation rules · Confidence scoring and routing uncertain extractions to a person	Practise
Stage 3	Time series basics: trend, seasonality and a baseline you must beat · Forecasting cash and demand with prediction intervals, not point guesses · RAG over tax circulars, contracts and internal finance policy	Build
Stage 4	Agents with tool calling across sheets, mail and accounting exports · n8n scheduling, audit trails and immutable run logs · Financial data confidentiality and what never goes to a public model	Harden and hand over

BY THE END YOU CAN

- Reconcile in minutes what currently takes days, with an audit trail
- Forecast with stated uncertainty instead of a single number
- Never present a figure you cannot trace

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- An extraction agent turning one recurring document type into validated structured rows
- A reconciliation notebook in Python that matches two sources and lists exceptions
- A scheduled MIS pack with every figure traceable to its source

07

Applied AI for Sales & Marketing

Research, content and pipeline agents, plus the data work behind knowing which lead is worth calling.

4 weeks · 24 hours

Online live + labs · 8 seats

Beginner to intermediate

Curriculum modules 01, 03, 04, 06, 07, 08, 09, 12

8 seats

WHO THIS IS FOR

- Sales executives and business development staff
- Marketing teams of one or two
- Founders doing their own selling and positioning

PREREQUISITE

Bring your CRM export and last quarter's campaigns.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	The AI taxonomy, and what “AI-powered” means on a competitor's site · Python and pandas on your own CRM export · Lead scoring as a classification problem: features, labels, evaluation	Understand
Stage 2	Precision versus recall when chasing leads costs time · Segmentation with clustering, and what it can and cannot tell you · Embeddings for similarity: finding accounts like your best ones	Practise
Stage 3	System prompts and schemas for outreach that is not obviously generated · Generative media: images, short video, voice, and the licensing rules · RAG over your product documentation, pricing and case studies	Build
Stage 4	Agents with tool calling: research, draft, log, follow up · n8n across CRM, mail, calendar and sheets · Attribution basics and measuring whether any of it worked	Harden and hand over

BY THE END YOU CAN

- Score and prioritise a pipeline with evidence
- Produce campaign material in-house without an agency
- Run outreach agents that stay specific enough to earn replies

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A research-and-outreach agent that prepares a briefing and a first draft per prospect
- A lead-scoring model on your own historical pipeline, honestly evaluated
- A follow-up workflow that drafts and queues for approval, never sends blind

08 Applied AI for Students

The technical foundation employers actually test for, ending in an agent and a project you can defend.

6 weeks · 36 hours

Online live + labs · 8 seats

Beginner to intermediate

Curriculum modules 01, 02, 03, 04, 05, 07, 08, 12, 13

8 seats

WHO THIS IS FOR

- School seniors, undergraduates and postgraduates
- Students building project, thesis or portfolio work
- Anyone entering a job market that assumes AI fluency

PREREQUISITE

None. Expect real homework.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	AI, ML, DL, GenAI and the vocabulary of every interview · Python properly: types, functions, classes, files, errors, Git · NumPy and pandas on real datasets, not toy ones	Understand
Stage 2	The mathematics that matters: vectors, gradients, probability · Supervised learning: regression, classification, train/test, leakage · Overfitting, cross-validation, bias-variance and regularisation	Practise
Stage 3	Neural network fundamentals and backpropagation, built once by hand · Tokens, embeddings, attention and how an LLM produces text · Prompt engineering with schemas, few-shot and evaluation	Build
Stage 4	RAG: chunking, embedding, retrieval, citation · Agents: tool calling, memory, loop control, tracing · Publishing work: repository, README, evaluation, demonstration	Harden and hand over

BY THE END YOU CAN

- Answer technical interview questions with your own code as evidence
- Build, evaluate and document a model without supervision
- Show a portfolio of working systems instead of a certificate list

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A machine learning project on a real dataset with a full evaluation report
- A RAG study assistant over your own course material, with citations
- An agent that automates one part of your academic or project workflow

09

Applied AI for Government & Compliance

Retrieval and drafting agents for document-heavy statutory work, with human sign-off designed into the architecture.

4 weeks · 24 hours

On site + online labs · 8 seats

Beginner to intermediate

Curriculum modules 01, 03, 04, 07, 08, 09, 12, 13, 33

8 seats

WHO THIS IS FOR

- Compliance, legal support and regulatory affairs staff
- Public-sector and PSU administrative teams
- Consultants filing on behalf of clients

PREREQUISITE

Bring your rule set, circulars and one filing process.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	The AI taxonomy and what may be represented to a regulator · Tokens, context limits and why long acts must be chunked · Python and pandas for obligation registers and deadline tracking	Understand
Stage 2	OCR and layout parsing of scanned notifications and orders · Structured extraction with validation against a schema · Embeddings and hybrid search across acts, rules and circulars	Practise
Stage 3	Reranking and citation binding so every answer is traceable · Refusal behaviour: what the system must say when it does not know · Agents with tool calling and mandatory human approval gates	Build
Stage 4	n8n for deadline monitoring, escalation and evidence capture · Audit logging, versioning and demonstrability to an inspector · DPDP and record-keeping obligations in practice	Harden and hand over

BY THE END YOU CAN

- Locate the governing provision in seconds, with the source
- Draft filings faster while keeping the signature human
- Show an inspector precisely how an output was produced

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A retrieval agent over your own statutory corpus that answers with the exact provision
- A drafting workflow with a mandatory review gate before anything is filed
- An obligation tracker that escalates before a deadline is missed

B

Core skill courses

Two to six weeks, one capability each, taught by building. These are for people who want to be able to do the thing themselves rather than be shown it.

COURSES IN THIS FAMILY

[10 · Prompt Engineering](#)[11 · AI Automation with n8n](#)[12 · Python for AI](#)[13 · Data Analysis with AI](#)[14 · Machine Learning](#)[15 · Deep Learning](#)[16 · Natural Language Processing](#)[17 · Computer Vision](#)[18 · Generative Media](#)

10

Prompt Engineering

Turning ad-hoc chatting into versioned, tested assets a whole team can run.

2 weeks · 12 hours

Online live · 8 seats

Intermediate

Curriculum modules 08

8 seats

WHO THIS IS FOR

- Anyone whose prompts work once and then stop working
- Teams standardising how they use AI
- Builders who need reliable structured output

PREREQUISITE

Comfortable using a chat interface.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Role, context, task, constraints, examples · Output schemas and JSON-shaped responses	Understand
Stage 2	Zero-shot, few-shot and worked examples · Chain of thought and decomposition	Practise
Stage 3	Self-critique and revision passes · Building a test set of hard inputs	Build
Stage 4	Measuring and tracking failure rate · Prompt injection and defences	Harden and hand over

BY THE END YOU CAN

- Write instructions a colleague can run without you
- Prove a prompt works instead of asserting it
- Harden a prompt against hostile or messy input

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

11 · AI Automation with n8n

18 · Generative Media

19 · AI Agents — Foundations

+1 more

What you build on this course

- A prompt asset with a 20-input test log and a stated failure rate
- A team prompt library with versions and owners
- An adversarial test that breaks somebody else's prompt

11

AI Automation with n8n

Work that runs on a trigger instead of on somebody remembering.

3 weeks · 18 hours

Online live · 8 seats

Intermediate

Curriculum modules 09

8 seats

WHO THIS IS FOR

- Operations and admin staff drowning in repetition
- Small businesses without a developer
- Consultants automating client processes

PREREQUISITE

Course 10 or equivalent.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Trigger, steps, output, human checkpoint · n8n nodes, credentials and connections	Understand
Stage 2	Webhooks and scheduled runs · Extraction from documents and email	Practise
Stage 3	Confidence thresholds and routing · Error handling, retries and alerts	Build
Stage 4	Run logs and cost per execution · Testing on copies, never on production	Harden and hand over

BY THE END YOU CAN

- Remove a recurring manual task permanently
- Recover cleanly when a workflow breaks
- Hand the workflow to a colleague with an SOP

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

20 · Agentic Automation for Business

What you build on this course

- One workflow running unattended on your own data
- An approval gate for low-confidence cases
- A run log and failure alert for the whole thing

12

Python for AI

The working environment every later course assumes — Python, data, Git and APIs.

4 weeks · 24 hours

Online live · 8 seats

Intermediate

Curriculum modules 03

8 seats

WHO THIS IS FOR

- Beginners moving from tools to building
- Analysts outgrowing spreadsheets
- Anyone heading toward the engineering courses

PREREQUISITE

None, but expect homework.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Types, functions, classes, files, errors · NumPy arrays and vectorisation	Understand
Stage 2	Pandas: load, clean, reshape, group · Plotting and quick visual checks	Practise
Stage 3	Virtual environments and packages · Git, GitHub, branches and reviews	Build
Stage 4	Calling APIs and parsing JSON · Structuring a project others can run	Harden and hand over

BY THE END YOU CAN

- Read an error and fix it without panic
- Handle data that breaks Excel
- Ship code somebody else can run

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

14 · Machine Learning

19 · AI Agents — Foundations

21 · Agent Engineering

+5 more

What you build on this course

- A cleaned and analysed real dataset, end to end
- A command-line tool that calls a live API
- A documented public repository

13 Data Analysis with AI

From messy operational data to a defensible conclusion, with the arithmetic verifiable.

3 weeks · 18 hours

Online live · 8 seats

Intermediate

Curriculum modules 03, 04

8 seats

WHO THIS IS FOR

- Analysts and MIS staff
- Operations and quality teams sitting on unused data
- Managers who receive numbers and must trust them

PREREQUISITE

Excel fluency; Python helpful.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Profiling a dataset before believing it · Cleaning, joining and deduplicating	Understand
Stage 2	Descriptive statistics that matter · Segmentation and cohort views	Practise
Stage 3	Trend and seasonality basics · Charts that do not mislead	Build
Stage 4	Written commentary from verified figures · Where AI helps and where it invents	Harden and hand over

BY THE END YOU CAN

- Turn a raw export into a decision-ready summary
- Spot the data problem before the meeting
- Defend every number you present

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A reproducible analysis of your own operational data
- A monthly dashboard or report pack
- A written commentary with every figure traceable

14 Machine Learning

The classical toolkit that still beats deep learning on most real tabular problems.

6 weeks · 36 hours

Online live · 8 seats

Intermediate

Curriculum modules 02, 04

8 seats

WHO THIS IS FOR

- Engineers and analysts moving into modelling
- Students building a serious portfolio
- Anyone who must evaluate a data science vendor

PREREQUISITE

Course 12 or equivalent Python.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Regression: linear, ridge, lasso · Classification: logistic, kNN, SVM, Naive Bayes	Understand
Stage 2	Trees, Random Forest, XGBoost, LightGBM · Clustering: k-means, DBSCAN, hierarchical	Practise
Stage 3	PCA, t-SNE and dimensionality reduction · Cross-validation and leakage	Build
Stage 4	Bias-variance, regularisation, tuning · Metrics beyond accuracy	Harden and hand over

BY THE END YOU CAN

- Choose the right algorithm and justify it
- Detect overfitting and leakage before deployment
- Report performance honestly to a non-technical room

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

15 · Deep Learning

16 · Natural Language Processing

32 · Time Series & Forecasting

What you build on this course

- A predictive model on real operational data with a full evaluation report
- A classification model with the metric chosen to match business risk
- A segmentation with an actual recommendation attached

15

Deep Learning

Neural networks built by hand, then in PyTorch, until the architecture stops being a mystery.

6 weeks · 36 hours

Online live · 8 seats

Advanced

Curriculum modules 05

8 seats

WHO THIS IS FOR

- Engineers heading into vision, NLP or LLM work
- Students targeting research or ML roles
- Anyone who needs to reason about model internals

PREREQUISITE

Course 14.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Perceptron to multi-layer network · Backpropagation, step by step	Understand
Stage 2	Activations, initialisation, optimisers · Dropout, batch norm, regularisation	Practise
Stage 3	CNNs, pooling, ResNet and skip connections · RNN, LSTM, GRU and their limits	Build
Stage 4	Attention and the Transformer block · Transfer learning and fine-tuning	Harden and hand over

BY THE END YOU CAN

- Explain what every layer is doing
- Read a training curve and diagnose it
- Decide when deep learning is the wrong choice

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

17 · Computer Vision

26 · LLM Engineering

What you build on this course

- A neural network from scratch in NumPy, including the backward pass
- A CNN trained on your own images
- A fine-tuned pretrained model with the gain measured

16

Natural Language Processing

Language as data — classification, extraction and similarity, including the cheap methods that still win.

5 weeks · 30 hours

Online live · 8 seats

Advanced

Curriculum modules 10

8 seats

WHO THIS IS FOR

- Teams processing large volumes of text
- Engineers building document pipelines
- Anyone paying per row for what a classifier could do

PREREQUISITE

Course 14.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Tokenisation, normalisation, n-grams · TF-IDF and classical baselines	Understand
Stage 2	word2vec, GloVe, fastText · Contextual and sentence embeddings	Practise
Stage 3	Classification and entity recognition · Topic modelling and clustering	Build
Stage 4	Summarisation and translation · Precision, recall, F1 and error analysis	Harden and hand over

BY THE END YOU CAN

- Pick the cheapest method that meets the accuracy bar
- Extract structured fields from unstructured text
- Evaluate a language system properly

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A classifier on your own documents, benchmarked against an LLM on cost and accuracy
- An entity extractor for one real document type
- Semantic search over a folder of reports

17

Computer Vision

Classification, detection, segmentation and OCR — with the deployment reality of a real camera feed.

5 weeks · 30 hours

Online live · 8 seats

Advanced

Curriculum modules 11

8 seats

WHO THIS IS FOR

- Manufacturing and safety teams with camera infrastructure
- Engineers building inspection systems
- Teams digitising paper records at scale

PREREQUISITE

Course 15.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Image basics, augmentation, annotation · Classification and transfer learning	Understand
Stage 2	Detection: YOLO and R-CNN families · Segmentation: U-Net, Mask R-CNN, SAM	Practise
Stage 3	OCR, layout analysis, table extraction · Tracking across frames	Build
Stage 4	Edge versus server inference · Lighting, occlusion, privacy and consent	Harden and hand over

BY THE END YOU CAN

- Choose the lightest task that answers the question
- Report detection performance honestly
- Anticipate why the pilot will fail on the real line

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A PPE or defect detection prototype with its false-positive rate measured
- A scanned register digitised into structured data
- A cost and frame-rate comparison for two deployment options

18

Generative Media

Image, video, voice and 3D generation, with the licensing and disclosure questions treated seriously.

3 weeks · 18 hours

Online live · 8 seats

Intermediate

Curriculum modules 06

8 seats

WHO THIS IS FOR

- Marketing, training and communication teams
- Schools and companies producing their own material
- Anyone commissioning visual work regularly

PREREQUISITE

Course 10 helpful.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	How diffusion actually works · Text-to-image and image-to-image	Understand
Stage 2	Inpainting, conditioning and control · Style consistency across a set	Practise
Stage 3	Text-to-video and frame consistency · Voice synthesis and bilingual narration	Build
Stage 4	Licensing and commercial rights · Deepfakes, consent and watermarking	Harden and hand over

BY THE END YOU CAN

- Produce publication-quality material in-house
- Answer a client's question about rights correctly
- Keep a consistent look across a whole campaign

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A branded visual set produced from a written brief
- A bilingual narrated training video, end to end
- A style guide that keeps output consistent

C

The agent track

Six courses on systems that decide their own next step. This is where the field is moving fastest and where settled curriculum does not yet exist, which is exactly why it is worth teaching properly.

COURSES IN THIS FAMILY

[19 · AI Agents — Foundations](#)[20 · Agentic Automation for Business](#)[21 · Agent Engineering](#)[22 · Agent Evaluation & Guardrails](#)[23 · Agent Operations in Production](#)[24 · Build an Agent Product](#)

19 AI Agents — Foundations

The first working agent: a system that decides its own next step, calls a tool, and knows when to stop.

3 weeks · 18 hours

Online live · 8 seats

Intermediate

Curriculum modules 12

8 seats

WHO THIS IS FOR

- Builders moving beyond single prompts
- Automation users hitting the limits of fixed workflows
- Anyone asked to “build an agent” without a definition of one

PREREQUISITE

Course 10; Course 12 helpful.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	What an agent is, and what is just a workflow · The reason, act, observe loop	Understand
Stage 2	Tool and function calling schemas · Designing a tool the model can use correctly	Practise
Stage 3	Short-term context and working memory · Termination conditions and loop caps	Build
Stage 4	Reading a trace to see what it did · Where an agent is the wrong answer	Harden and hand over

BY THE END YOU CAN

- Tell an agent problem from a workflow problem
- Write tools an agent uses correctly rather than creatively
- Explain every step your agent took

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

20 · Agentic Automation for Business

21 · Agent Engineering

22 · Agent Evaluation & Guardrails

+1 more

What you build on this course

- A single-tool agent that completes a real task end to end
- A three-tool agent with a stated stopping rule
- A trace review of your own agent's decisions

20

Agentic Automation for Business

Agents placed inside real business systems — mail, sheets, drives, tickets — with approval gates that hold.

4 weeks · 24 hours

Online live · 8 seats

Intermediate

Curriculum modules 09, 12

8 seats

WHO THIS IS FOR

- Operations leads automating judgement-heavy work
- Consultants delivering automation to clients
- Teams whose processes are too variable for fixed rules

PREREQUISITE

Courses 11 and 19.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Where judgement beats a decision tree · Agents inside n8n and existing workflows	Understand
Stage 2	Connecting to mail, sheets, drives and tickets · Approval gates and escalation paths	Practise
Stage 3	Confidence scoring and routing · Handling exceptions without stalling	Build
Stage 4	Cost per run and volume planning · Change management with the people affected	Harden and hand over

BY THE END YOU CAN

- Automate work that rules alone could never cover
- Keep a named human accountable for every output
- Predict what it will cost before you scale it

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- An agent handling one judgement-heavy process with a human approval gate
- An exception path that escalates rather than guesses
- A cost model for running it at real volume

21 Agent Engineering

The deep version — planning, multi-agent structures, durable memory, and tool design that survives contact.

8 weeks · 48 hours

Online live · 8 seats

Advanced

Curriculum modules 12, 13, 14

8 seats

WHO THIS IS FOR

- Developers building agent products
- Engineers moving from prototypes to systems
- Technical founders shipping agentic features

PREREQUISITE

Courses 12 and 19.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Planning: ReAct, plan-and-execute, reflection · Supervisor and worker architectures	Understand
Stage 2	Sequential, parallel and branching pipelines · Long-term memory and retrieval over history	Practise
Stage 3	Model Context Protocol and tool servers · State, checkpoints and resumability	Build
Stage 4	Concurrency, queues and long-running tasks · Structured output and schema enforcement	Harden and hand over

BY THE END YOU CAN

- Design an agent architecture and defend the choices
- Build systems that recover instead of collapsing
- Expose internal systems to an agent without opening the door

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

23 · Agent Operations in Production

24 · Build an Agent Product

What you build on this course

- A multi-agent system where one agent drafts and another critiques
- A durable agent that survives restart and resumes work
- A tool server exposing your own systems safely

22

Agent Evaluation & Guardrails

Proving an agent works, and constraining what it can do when it does not.

4 weeks · 24 hours

Online live · 8 seats

Advanced

Curriculum modules 12, 08

8 seats

WHO THIS IS FOR

- Teams with an agent already running and no evidence it works
- Engineers responsible for AI that touches customers
- Anyone signing off on autonomous behaviour

PREREQUISITE

Course 19.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Building an evaluation set for agent behaviour · Task success versus step correctness	Understand
Stage 2	Failure taxonomies and trace analysis · LLM-as-judge and its limits	Practise
Stage 3	Permission scoping per tool · Loop, spend and rate caps	Build
Stage 4	Prompt injection through tool output · Audit logging every action taken	Harden and hand over

BY THE END YOU CAN

- State your agent's success rate with evidence
- Bound the damage an agent can do
- Show an auditor exactly what it did and why

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

23 · Agent Operations in Production

What you build on this course

- An evaluation harness scoring your agent across 30 scenarios
- A guardrail layer with permission scoping and hard caps
- An injection test suite that tries to hijack your own agent

23

Agent Operations in Production

Keeping autonomous systems alive, affordable and accountable once real people depend on them.

4 weeks · 24 hours

Online live · 8 seats

Advanced

Curriculum modules 12, 15

8 seats

WHO THIS IS FOR

- Teams running agents in production
- Engineers owning uptime and spend
- Consultants handing systems to client IT

PREREQUISITE

Courses 21 or 22.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Deployment patterns for long-running agents · Monitoring: latency, cost, failures, quality	Understand
Stage 2	Runaway loop detection and kill switches · Spend alerts and per-tenant budgets	Practise
Stage 3	Versioning agents and rolling back · Incident response for autonomous systems	Build
Stage 4	Upstream model changes and deprecations · Runbooks and handover	Harden and hand over

BY THE END YOU CAN

- Detect a misbehaving agent before the client does
- Keep monthly spend inside a stated ceiling
- Hand the system over without handing over yourself

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A monitored agent deployment with alerting
- A kill switch and spend cap proven by test
- An incident runbook a colleague can follow at 2am

24

Build an Agent Product

A multi-tenant agent SaaS built end to end — the shape of a real product, not a demonstration.

8 weeks · 48 hours

Online live · 8 seats

Advanced

Curriculum modules 12, 13, 14, 15

8 seats

WHO THIS IS FOR

- Founders building an AI product
- Engineers taking an internal tool to market
- Teams serving several client organisations from one system

PREREQUISITE

Courses 21 and 25.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Multi-tenant data isolation · Per-tenant configuration and knowledge	Understand
Stage 2	Onboarding a new client organisation · Usage metering and billing hooks	Practise
Stage 3	Role-based access and admin views · Support, feedback and improvement loops	Build
Stage 4	Cost per tenant and pricing structure · Security review before first customer	Harden and hand over

BY THE END YOU CAN

- Ship something a paying customer could use
- Prove one tenant cannot see another's data
- Price it from real cost rather than hope

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A working multi-tenant agent product with two isolated tenants
- An admin console showing usage and cost per tenant
- A security and data-isolation review document

D

Advanced & specialist

Four to ten weeks, senior level. Retrieval, backend, operations, evaluation, security and governance — the work that decides whether an AI system survives its first year.

COURSES IN THIS FAMILY

25 · RAG Engineering

26 · LLM Engineering

27 · AI Backend & APIs

28 · MLOps & Deployment

29 · AI Evaluation Engineering

30 · AI Security

31 · Document Intelligence

32 · Time Series & Forecasting

33 · AI Governance for Indian Companies

34 · AI Product Management



25

RAG Engineering

Answers grounded in your own documents, with citations, permissions and a measured accuracy.

6 weeks · 36 hours

Online live · 8 seats

Advanced

Curriculum modules 13

8 seats

WHO THIS IS FOR

- Teams building internal knowledge assistants
- Engineers whose first RAG demo disappointed
- Anyone answering questions from a large document estate

PREREQUISITE

Course 12.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Parsing, chunking and overlap strategy · Embedding model selection and cost	Understand
Stage 2	Vector stores: FAISS, Chroma, pgvector, Qdrant · Hybrid search with BM25	Practise
Stage 3	Cross-encoder reranking · Permission filtering per department	Build
Stage 4	Citation binding and refusal behaviour · Retrieval recall, faithfulness, drift	Harden and hand over

BY THE END YOU CAN

- State your system's retrieval and faithfulness scores
- Improve results deliberately rather than by guessing
- Refuse cleanly when the documents do not answer

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

24 · Build an Agent Product

What you build on this course

- A RAG system over your own estate with mandatory citations
- A scored evaluation across 30 real questions
- Permission filtering proven with two user roles

26

LLM Engineering

Working with models as engineering components — selection, control, adaptation and self-hosting.

10 weeks · 60 hours

Online live · 8 seats

Advanced

Curriculum modules 07, 13, 14

8 seats

WHO THIS IS FOR

- Developers building on model APIs
- Teams weighing hosted against open models
- Engineers who need control over cost and behaviour

PREREQUISITE

Courses 12 and 15.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Tokenisation, context, attention in practice · Decoding: temperature, top-p, penalties	Understand
Stage 2	Structured output and schema enforcement · Prompt caching and cost engineering	Practise
Stage 3	Fine-tuning: when it helps and when it does not · LoRA and parameter-efficient adaptation	Build
Stage 4	Quantisation and local inference · Open model selection and self-hosting	Harden and hand over

BY THE END YOU CAN

- Choose a model on evidence, not marketing
- Cut inference cost without losing quality
- Run a model on your own infrastructure when policy demands it

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A model comparison with your own evaluation set and cost figures
- A fine-tuned or adapted model with the gain measured against the base
- A self-hosted deployment serving a real workload

27

AI Backend & APIs

The layer that turns a working notebook into something other people and systems can call.

5 weeks · 30 hours

Online live · 8 seats

Advanced

Curriculum modules 14

8 seats

WHO THIS IS FOR

- Developers productionising AI prototypes
- Full-stack engineers adding AI features
- Founders building the first version themselves

PREREQUISITE

Course 12.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	REST design and FastAPI endpoints · Validation and typed request models	Understand
Stage 2	Authentication and key management · Streaming responses to a client	Practise
Stage 3	Postgres and Supabase schema design · Storing records and embeddings together	Build
Stage 4	Caching, queues and background jobs · Rate limiting and abuse prevention	Harden and hand over

BY THE END YOU CAN

- Serve AI to a department without exposing keys
- Keep response times acceptable under real load
- Meter usage so the service can be billed

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

28 · MLOps & Deployment

30 · AI Security

What you build on this course

- An authenticated API serving a model with streaming
- A schema holding records, embeddings and audit logs
- A load test showing behaviour at ten times current volume

28

MLOps & Deployment

Shipping, monitoring and maintaining systems after the interesting part is finished.

6 weeks · 36 hours

Online live · 8 seats

Advanced

Curriculum modules 15

8 seats

WHO THIS IS FOR

- Engineers who have deployed once and been burned
- Teams inheriting systems they did not build
- Consultants who must hand over cleanly

PREREQUISITE

Courses 12 and 27.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Docker images and reproducible builds · CI pipelines and automated tests	Understand
Stage 2	Deploy targets and rollback strategy · Secrets, config and environments	Practise
Stage 3	Latency, cost, error and quality monitoring · Data and concept drift detection	Build
Stage 4	Scheduled re-evaluation · Runbooks, ownership and handover	Harden and hand over

BY THE END YOU CAN

- Deploy without ceremony or fear
- Catch quality decay before a user reports it
- Hand a system over with a runbook that works

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A deployment pipeline that ships on push and rolls back cleanly
- A monitoring dashboard covering the four production numbers
- A drift check that runs on a schedule and alerts

29

AI Evaluation Engineering

The discipline that separates a system you can trust from one that merely demonstrated well.

4 weeks · 24 hours

Online live · 8 seats

Advanced

Curriculum modules 07, 13

8 seats

WHO THIS IS FOR

- Teams shipping AI features without a quality bar
- Engineers asked “how do you know it works?”
- Product owners defining acceptance criteria

PREREQUISITE

Course 10.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Designing a golden question set · Choosing metrics that match the business risk	Understand
Stage 2	LLM-as-judge and where it misleads · Human review protocols and sampling	Practise
Stage 3	Regression testing after every change · A/B evaluation in production	Build
Stage 4	Failure taxonomies and error analysis · Reporting quality to non-technical stakeholders	Harden and hand over

BY THE END YOU CAN

- Answer the accuracy question with a number
- Stop shipping regressions unknowingly
- Set an acceptance bar before building

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A golden set and scoring harness for one live system
- A regression suite that runs on every change
- A quality report a manager can read

30

AI Security

Attacking your own systems before somebody else does — injection, leakage and abuse.

4 weeks · 24 hours

Online live · 8 seats

Advanced

Curriculum modules 08, 12, 14

8 seats

WHO THIS IS FOR

- Engineers exposing AI to untrusted input
- Security and compliance reviewers
- Teams handling confidential or personal data

PREREQUISITE

Courses 19 or 27.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Prompt injection, direct and indirect · Injection through retrieved documents and tools	Understand
Stage 2	Data leakage and context contamination · Output handling and downstream execution risk	Practise
Stage 3	Excessive agency and permission design · Rate limiting, abuse and cost attacks	Build
Stage 4	OWASP LLM Top 10 in practice · Red teaming your own system	Harden and hand over

BY THE END YOU CAN

- Break your own system before a stranger does
- Reduce blast radius when something gets through
- Pass a security review with evidence

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A red-team report against your own AI system
- A defence layer with input, output and permission controls
- A residual-risk statement a reviewer would accept

31

Document Intelligence

Turning scans, forms and registers into structured, validated data at volume.

5 weeks · 30 hours

Online live · 8 seats

Advanced

Curriculum modules 11, 13

8 seats

WHO THIS IS FOR

- Teams digitising paper archives
- Finance, compliance and records functions
- Anyone re-keying data that already exists on paper

PREREQUISITE

Course 12.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	OCR quality and preprocessing · Layout analysis and reading order	Understand
Stage 2	Table extraction from scans · Form and field extraction	Practise
Stage 3	Handwriting: what is realistic · Validation rules and confidence scoring	Build
Stage 4	Human review queues for low confidence · Volume, cost and throughput planning	Harden and hand over

BY THE END YOU CAN

- Digitise an archive without silently corrupting it
- Route uncertain extractions to a person
- Quote a realistic cost per thousand pages

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A pipeline turning one real document type into validated structured data
- A confidence-based review queue
- An accuracy report by field, not by document

32

Time Series & Forecasting

Demand, consumption and failure prediction, with uncertainty stated rather than hidden.

5 weeks · 30 hours

Online live · 8 seats

Advanced

Curriculum modules 04, 05

8 seats

WHO THIS IS FOR

- Planning, maintenance and energy teams
- Analysts asked for next quarter's numbers
- Manufacturing teams with sensor history

PREREQUISITE

Course 14.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Trend, seasonality and decomposition · Baselines you must beat	Understand
Stage 2	ARIMA and exponential smoothing · Gradient boosting for forecasting	Practise
Stage 3	Sequence models and when they help · Backtesting properly	Build
Stage 4	Prediction intervals and uncertainty · Predictive maintenance framing	Harden and hand over

BY THE END YOU CAN

- Forecast with honest uncertainty
- Prove the model beats guessing
- Explain a forecast to the people who must act on it

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A forecast that beats a naive baseline, with the margin stated
- A backtest across multiple periods
- A maintenance or demand model with intervals, not point guesses

33

AI Governance for Indian Companies

Policy, data protection and vendor assessment for organisations adopting AI seriously.

3 weeks · 18 hours

Online live · 8 seats

Intermediate

Curriculum modules 01, 08

8 seats

WHO THIS IS FOR

- Compliance, legal and risk functions
- IT and security leads writing policy
- Directors accountable for AI decisions

PREREQUISITE

None.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Writing an internal AI usage policy · DPDP obligations in practical terms	Understand
Stage 2	ISO 42001 and what certification involves · Data classification and handling rules	Practise
Stage 3	Vendor and model assessment questions · Records, logging and demonstrability	Build
Stage 4	Human oversight requirements · Incident and disclosure procedure	Harden and hand over

BY THE END YOU CAN

- Adopt AI without creating a compliance problem
- Assess a vendor on substance
- Show a regulator or customer how you govern it

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- An AI usage policy tailored to your organisation
- A vendor assessment questionnaire
- A data-handling matrix by classification

34

AI Product Management

Scoping, pricing and roadmapping products whose behaviour is probabilistic rather than specified.

4 weeks · 24 hours

Online live · 8 seats

Intermediate

Curriculum modules 01, 07, 29

8 seats

WHO THIS IS FOR

- Product managers adding AI features
- Founders defining an AI product
- Consultants scoping AI engagements

PREREQUISITE

None.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Which problems suit probabilistic systems · Writing acceptance criteria without determinism	Understand
Stage 2	Evaluation-driven roadmaps · Cost modelling per user and per action	Practise
Stage 3	Latency and quality trade-offs · Failure UX and graceful degradation	Build
Stage 4	Pricing when unit cost varies · Communicating limits to customers	Harden and hand over

BY THE END YOU CAN

- Scope AI work that can actually be delivered
- Price a feature that costs money per use
- Say no to demands that probability cannot meet

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A product spec with evaluation-based acceptance criteria
- A unit-economics model for one AI feature
- A failure-mode and degradation plan

E

Flagship programmes

The full-length programmes. Every one is capped at eight participants, because above that the instructor cannot look over each person's shoulder while they build.

COURSES IN THIS FAMILY

35 · Applied AI Build Cohort

36 · AI Career Track

37 · LLM & Agent Engineering Track

38 · Corporate Custom Programme



35

Applied AI Build Cohort

Eight people, four weeks, one automated task each — the fastest route from curiosity to something running.

4 weeks · 16 hours live

Online live · 8 seats

All levels

Curriculum modules 01, 07, 08, 09, 13

8 seats

WHO THIS IS FOR

- Working professionals with a repetitive task
- Teams sending two or three people together
- Anyone who has taken courses and built nothing

PREREQUISITE

Bring one real recurring task. No task, no seat.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Baselining your current process in minutes and steps · Prompt assets for your own documents	Understand
Stage 2	Your own messy data, cleaned and analysed · Retrieval over your own standards	Practise
Stage 3	One workflow automated end to end · Human checkpoints and governance	Build
Stage 4	Hardening and error handling · Handover so a colleague can run it	Harden and hand over

BY THE END YOU CAN

- Cut a recurring task and prove the saving
- Run and maintain what you built without the instructor
- Present the result to your own management

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- One automation running unattended on your own accounts
- A one-page SOP a colleague can follow
- A measured before-and-after on a real task

36

AI Career Track

The full curriculum, all sixteen modules, ending in a capstone that is defended live.

6 months · 480 hours

Online live + labs · 8 seats

Beginner to advanced

Curriculum modules 01–16 complete

8 seats

WHO THIS IS FOR

- Career changers entering AI seriously
- Graduates who need built work, not certificates
- Professionals moving into an AI role internally

PREREQUISITE

Commitment, not qualifications.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Foundations: AI, mathematics, Python, ML, DL · Applications: GenAI, LLMs, prompting, automation	Understand
Stage 2	Engineering: NLP, vision, agents · Production: RAG, APIs, deployment	Practise
Stage 3	Evaluation throughout, not at the end · Weekly labs on real data	Build
Stage 4	Portfolio building from module three onward · Capstone scoping, build, defence	Harden and hand over

BY THE END YOU CAN

- Enter interviews with systems, not certificates
- Design, build, evaluate and deploy independently
- Hold your own with engineers who did a degree

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- Eight to ten portfolio projects across the curriculum
- One deployed capstone system with evaluation results
- A public repository and a defended demonstration

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LLM & Agent Engineering Track

For people who already write code and want to build agentic and retrieval systems properly.

12 weeks · 84 hours

Online live · 8 seats

Advanced

Curriculum modules 03, 05, 07, 12, 13, 14, 15

8 seats

WHO THIS IS FOR

- Software engineers moving into AI roles
- Backend developers adding AI to products
- Technical founders building the first version

PREREQUISITE

Working developer.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Model internals that affect engineering choices · Structured output and tool calling	Understand
Stage 2	RAG with hybrid search and reranking · Agent architectures and durable state	Practise
Stage 3	Evaluation harnesses and regression suites · Backend, auth, multi-tenancy	Build
Stage 4	Deployment, monitoring, cost control · Security and injection defence	Harden and hand over

BY THE END YOU CAN

- Build AI systems that survive real users
- Prove quality with a harness rather than a demo
- Take an AI engineering role or ship your own product

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- A production-grade RAG service with evaluation
- A multi-agent system with guardrails and tracing
- A deployed, monitored, cost-capped application

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Corporate Custom Programme

Modules chosen around one organisation's actual workflows, taught on that organisation's own data.

Scoped per client

On site or hybrid · 8 per batch

Mixed

Curriculum modules Selected per client

8 seats

WHO THIS IS FOR

- Companies training a whole department
- Plants and schools with site-specific processes
- Organisations that need policy and capability together

PREREQUISITE

A scoping session with the department head.

WHAT YOU WILL LEARN, AND IN WHAT ORDER

Stage	Covered	Mode
Stage 1	Scoping session: processes, data, constraints · Modules selected from the full curriculum	Understand
Stage 2	All examples drawn from the client's own work · Bilingual delivery where the floor needs it	Practise
Stage 3	Internal AI usage policy drafted alongside · Data-handling rules agreed before delivery	Build
Stage 4	Multiple batches of eight for larger teams · Post-programme review at thirty days	Harden and hand over

BY THE END YOU CAN

- Build capability in-house instead of renting it
- Standardise how the organisation uses AI
- Retain the tools, the skills and the policy

ASSESSMENT & CERTIFICATE

A quiz, one graded exercise on your own data, and the artefact below reviewed against a checklist. The certificate names what you built and the result you measured.

LEADS ON TO

Any course you meet the prerequisites for

What you build on this course

- Working internal tools built by the client's own staff
- An organisation-specific AI usage policy
- A capability map showing who can now do what

How to enrol, and what we will not claim

Not every course runs every month. Batches open when eight suitable people are ready, because a half-filled cohort of mismatched levels serves nobody.

HOW ENROLMENT WORKS

- A short call: what you do, what you would bring, whether this course fits
- Dates, tooling access and data-handling rules agreed in writing
- Payment before session one; full refund if you withdraw before session two
- Recordings for missed sessions; repeated absence means no certificate

WHAT THE CERTIFICATE IS, AND IS NOT

A private professional certificate naming what you built and the result you measured, verifiable online by anyone who asks. It carries the weight of evidence, not of accreditation.

Not a degree, diploma or government-recognised qualification, and not represented as one.

SUGGESTED ROUTES THROUGH THE CATALOGUE

If you are	Route	Ends with
A working professional	01 or 03 → 10 → 11 → 35	One task automated and running on your own accounts
A manager or owner	02 → 33 → 38	A policy, a pilot and a trained department
A student or fresher	08 → 12 → 36	A portfolio and a defended capstone
A developer	19 → 25 → 21 → 37	A deployed, evaluated, guarded agent system
A data person	13 → 14 → 32 or 16	Models in production with honest evaluation

Enquiries

- starsynai.com · Bangalore · Coimbatore · Nilgiris
- Corporate programmes are scoped directly with the department head, on site.
- Every course is capped at eight seats. Selection is by conversation, not by payment.